

Shan Yang

Senior Applied Scientist | Video Generation, Multimodal Modeling, GenAI Systems
alexyangshan@gmail.com | shanyang.me | [Google Scholar](https://scholar.google.com/citations?user=shanyang)

EXPERIENCE

Amazon Prime Video – Senior Applied Scientist Tech Lead | June 2025 – present

Lead scientist for **Prime Video's GenAI Live Action Studio**, building core models and pipelines for next-generation creative production tools * **Established Prime Video's post-training pipeline** (LoRA, ControlNet, Test-Time-Training, Self-forcing, Diffusion-forcing) enabling controllable, consistent, and context-preserving **generative video**, forming the adaptation layer for future GenAI creative systems * Driving foundational research in conditional video generation, advancing auto-regressive diffusion, test-time conditioning mechanisms, and controllable keyframed video synthesis using diffusion forcing * **Author of multiple internal publications** shaping the long-term GenAI video roadmap for Prime Video.

Amazon Search – Senior Applied Scientist Tech Lead | Sept 2021 - June 2025

Led and launched Amazon Video Search on the Amazon product search page, building a new multimodal customer experience adopted by millions of users * Developed the Video Search Ranker, combining video semantic representations with text, image, and product metadata features through a unified multimodal architecture optimized for large-scale retrieval and production search workloads * Architected the end-to-end Video Search system: retrieval, ranking, multimodal fusion, and inference * Led multimodal LLM efforts for captioning and VQA using **Schema-of-Thought** reasoning * Owned full-stack development, deployment, and A/B testing * Granted an Amazon patent for multimodal ranking innovations.

Google Research – Senior Research Software Engineer | Jun 2017 - Sept 2021

Led **multimodal video content creation** research, including **music-conditioned 3D dance generation** and **video-audio-motion modeling** * Conducted core **3D vision research**, focusing on **monocular video pose estimation** and structured motion understanding *

Served as the **primary owner and major contributor** for Google's **2D/3D reconstruction ML models**, covering architecture design, training, and evaluation * Built an **end-to-end monocular video analysis system** used for biological and biomechanical research * Major contributor to a Google patent in multimodal generative modeling and vision-based reconstruction.

Google Research, Mountain View — Research Intern | Summer 2016 (3 months)

3D Vision Research: Monocular Video 3D Human Pose Understanding * 3D Hair Reconstruction

Samsung Research America, San Jose — Software Engineer Intern | Summer 2014 (3 months) and Summer 2013 (3 months)

Advanced Processor Lab * Graphics hardware test infrastructure development

EDUCATION

University of North Carolina at Chapel Hill, NC — Ph.D. Computer Science | 2012 – 2017

Shanghai Jiao Tong University, China — B.E. Software Engineering | 2008 – 2012

PENDING PUBLICATIONS

SoT: Schema-of-Thought for Robust Video Question Answering, Xijun Wang, **Shan Yang**, 2024

We propose to apply the Schema Theory to dynamically prompt the multi-modal models to maximally reduce the poor perception performance problem. ChatGPT-4o with SoT achieved +13.3% improvement, Claude3.5 Sonnet with SoT achieved +28% improvement on MMVP. And we found SoT consistently improved the performance on public hallucination benchmarks.

Linear Recurrent Model for Test-Time Reasoning via Linear Quadratic Control, Peihao Wang, **Shan Yang**, 2025

We derive a novel neural layer that performs test-time planning to embed forward-looking computation directly within the model. We design a novel CUDA implementation that parallelizes this optimal planning on GPUs efficiently. By incorporating it as an adapter into pretrained LLMs, our approach enhances mathematical reasoning performance, compared with LLaMA3 8B, +35.8% on Math-500, +6.7% on AIME24, +10% on AIME25, +9.3% on AMC.

ClipDreamer: Towards Faithful and Visually-Grounded Product Video Generation, Zun Wang, **Shan Yang**, 2025

We propose to use the Test-Time-Training layer to inject video clip conditions to drive faithful motion and appearance product-centric video generation.

Teacher Diffusion Forcing: New Training Paradigm for Video Generation, Peihao Wang, **Shan Yang**, 2025

We introduce Teacher-Diffusion Forcing, a unified training paradigm that integrates the strengths of both teacher forcing and diffusion forcing to improve the efficiency and quality of video generation.

PUBLICATIONS

ViLA: Efficient video-language alignment for video question answering

Xijun Wang, Junbang Liang, Chun-Kai Wang, Kenan Deng, Yu Lou, Ming C Lin, **Shan Yang**, European Conference

on Computer Vision (ECCV), 2024

ICAR: Image-based Complementary Auto Reasoning

Xijun Wang, Anqi Liang, Junbang Liang, Ming Lin, Yu Lou, **Shan Yang**, AAAI 2023

MeSa: Masked, Geometric, and Supervised Pre-training for Monocular Depth Estimation

Muhammad Osama Khan, Junbang Liang, Chun-Kai Wang, **Shan Yang**, Yu Lou, NeurIPS 2023 Workshop SSLTheoryPractice

Attention Bottlenecks for Multimodal Fusion

Arsha Nagrani, **Shan Yang**, Anurag Arnab, Aren Jansen, Cordelia Schmid, Chen Sun, Neural Information Processing Systems (NIPS), 2021

Optical Mouse: 3D Mouse Pose From Single-view Video

Shan Yang*, Bo Hu*, David A. Ross, Avneesh Sud, Yi Liu, Graham Ruby, Bryan Seybold, Conference on Computer Vision and Pattern Recognition (CVPR), 2021 CV4Animal Workshop

Learn to Dance with AIST++: Music Conditioned 3D Dance Generation

Shan Yang*, Ruilong Li*, David A. Ross, Angjoo Kanazawa, International Conference on Computer Vision (ICCV), 2021

Physics-Inspired Garment Recovery from a Single-View Image

Shan Yang, Zherong Pan, Tanya Amert, Ke Wang, Licheng Yu, Tamara Berg, Ming C. Lin, ACM Transaction on Graphics (TOG), 2018

Learning-based Cloth Material Recovery from Video

Shan Yang, Junbang Liang, Ming C. Lin, International Conference on Computer Vision (ICCV), 2017

Modeling Context in Referring Expressions

Lichens Yu, Patric Poirson, **Shan Yang**, Alex Berg, Tamara Berg, European Conference on Computer Vision (ECCV), spotlight, 2016

Classification of Prostate Cancer Grades and T-Stages based on Tissue Elasticity Using Medical Image Analysis

Shan Yang, Vladimir Jojic, Jun Lian, Ronald Chen, Hongtu Zhu and Ming C. Lin, International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), 2016

Bayesian Estimation of Non-Rigid Mechanical Parameters Using Temporal Sequences of Deformation Samples

Shan Yang, Ming C. Lin, IEEE International Conference on Robotics and Automation (ICRA), 2016

Material Cloning: Acquiring Elasticity Parameters from Images

Shan Yang, Ming C. Lin, IEEE Transactions on Visualization and Graphics (TVCG), 2016

Simultaneous Estimation of Elasticity for Multiple Deformable Bodies

Shan Yang, Ming C. Lin, Journal Computer Animation and Virtual Worlds (CAVW), 2015

Patents

Monitoring Animal Pose Dynamics from Monocular Images, Bryan Andrew Seybold, **Shan Yang**, Bo Hu, Kevin Murphy, David Ross, 2022

Image-based Complementary Item Recommendation Algorithm, **Shan Yang**, Xijun Wang, Ming Lin, Anqi Liang, 2022

Auto Design: ML-based Furniture Layout Recommendation Algorithm, **Shan Yang**, Esra Boduroglu, Junbang Liang, Kenan Deng, Ramesh Krishnamurphy, Tomas Vicent, Yu Lou, 2022

VLAP: Efficient Video-Language Alignment via Frame Prompting and Temporal Distilling, **Shan Yang**, Xijun Wang, 2023